

Spatially blocked pluvial-flood susceptibility learning on the H3 grid using heterogeneous open public data

Highlights

- An H3-native framework links evidence assembly, spatially blocked validation, and resolution control on one hierarchical grid
 - Heterogeneous open sources are kept distinct as **evidence-positive** / **evidence-unrecorded** labels (model-derived DEP, crowd-reported 311, observed Ida high-water marks), not verified flood / non-flood ground truth
 - Prevalence-aware baselines change the reading of classification skill; the Lower Manhattan model (Option B, $n = 262$) exceeds always-positive on accuracy (0.824 vs 0.637) and F1 (0.861 vs 0.769)
 - Domain-masked native fine-resolution overlay with area-budget fractional hotspots shows substantial scale loss (mean hotspot Jaccard 0.136 to R8 at a 10% area budget; R10→R9 support unified at $n = 262$)
 - The fitted score is an H3 susceptibility score $S_h(c)=f_\theta(X_c)$; rainfall conditioning is deferred to event-based $S_h(c,r)$
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Abstract

Urban pluvial flooding develops rapidly when intense rainfall overwhelms drainage before water reaches a watercourse, and screening it at city scale requires representations that are computationally scalable, updateable as new evidence arrives, and evaluated with explicit control of spatial dependence. Many data-driven approaches rely on proprietary damage or insurance labels, and random train-test splits can inflate skill when spatially proximate observations appear in both training and test sets. This study evaluates an H3-native framework in which the hexagonal discrete global grid provides the common spatial support for evidence assembly, spatially blocked validation, and resolution control, using only heterogeneous open public data. Flood evidence is assembled on H3 cells from three sources that are kept distinct: model-derived stormwater polygons (New York City DEP, categories 1–2), crowd-reported street-flooding complaints (NYC 311), and observed high-water marks (USGS Hurricane Ida). Binary cell labels are framed as **evidence-positive** versus **evidence-unrecorded**, not as verified flood versus verified non-flood. Gradient-boosting classifiers and an evidence-score regressor are fitted and evaluated under H3-block spatial cross-validation in which entire parent cells are withheld; after evaluation, separate deployment models are refitted on all cells for descriptive maps and adaptive screening only. In the Lower Manhattan pilot (manuscript bbox, $n = 262$ R9 cells, 63.7% evidence-positive; Major Revision Option B), the model attains spatial cross-validation accuracy 0.824 ± 0.055 and F1 0.861, above the always-positive baseline (0.637 and 0.769), with pooled out-of-fold ROC-AUC 0.850 and average precision 0.851. In the larger pilot ($n = 956$ cells, 48.2% evidence-positive), accuracy is 0.819 ± 0.024 and F1 is 0.822, both above the fold-mean constant-class baselines, with pooled out-of-fold ROC-AUC 0.880 and average precision 0.815. A domain-masked native fine-resolution label overlay (R10 fine parents restricted to the modelling R9 set) with tie-aware fractional hotspot membership and a strict 10% area budget shows that hotspot membership degrades under coarsening (mean Jaccard

0.136 to R8; 0.220 to R9), and adaptive refinement concentrates representation into about 57% as many cells as a uniform fine grid. The primary screening quantity is the cell susceptibility score $S_h(c)=f_\theta(X_c)$; because training rainfall is constant, rainfall-conditioned $S_h(c,r)$ remains unevaluated and is deferred. The results illustrate a reproducible, prevalence-aware spatial-evaluation architecture for screening pluvial-flood susceptibility from open evidence, while citywide skill and rainfall-conditioned discrimination remain open questions. Production assembly is fail-closed: missing layers or NaNs abort rather than silently filling synthetic features.

Keywords: pluvial flood susceptibility; H3; discrete global grid; spatial cross-validation; machine learning; open data.

1. Introduction

Urban flooding occurs predominantly during intense rainfall in densely built areas with limited drainage, and its pluvial form—flooding that develops when rainfall overwhelms drainage before entering watercourses—can appear rapidly and with little warning [1]. Assessing this hazard at city scale requires representations that are computationally scalable, updateable as new observations arrive, and evaluated in a way that does not inflate performance. Two limitations recur in data-driven pluvial-flood screening. The first is that some published indices rely on proprietary damage or insurance records, so the underlying labels are neither public nor transferable to jurisdictions without comparable records. The second is evaluation: random train-test splits can overestimate generalisation when spatially proximate observations occur in both the training and test sets because of spatial autocorrelation. For city-scale screening these constraints create a spatial-representation problem as well as a modelling problem: observations, predictions, and evaluation need a spatial support that remains coherent as scale changes.

Discrete Global Grid Systems, and the hexagonal H3 system in particular [2], provide a scalable spatial substrate for integrating observations, predictions, and multi-resolution analysis. H3 is a hierarchical, predominantly hexagonal spatial index with neighbourhood operations and parent–child relationships across resolutions [3]. Hexagonal grids have been applied to multi-scale flood mapping under climate scenarios [4], and a closely related line of work aggregates a pre-existing machine-learning building-level pluvial susceptibility index into H3 cells to reduce query cost and expose resolution-dependent hotspot loss [5,6]. In that formulation H3 serves primarily as a multi-resolution aggregation and communication layer for an inherited index, leaving open the question of how the same hierarchy might support model fitting and evaluation across unseen spatial blocks.

These limitations motivate an H3-native learning and evaluation architecture in which the grid acts as the common spatial support rather than only as an aggregation layer. Heterogeneous public evidence is assembled directly on H3 cells, coarse H3 parents define the spatial holdouts, and the same hierarchy is subsequently used to examine scale effects and guide selective refinement. The contribution therefore lies in linking evidence construction, spatial validation, and resolution control within one hierarchical spatial reference; each of these components is established separately in prior work [4,7,8,9,10]. Throughout, the fitted positive-class model output is treated as a susceptibility score and is kept distinct from both feature-importance measures and the H3-aggregated building index of Svelling et al. [5].

Accordingly, the analysis asks whether predictive performance persists when entire H3 parent blocks are withheld, relative to class-prevalence baselines; how hotspot membership changes as fine-scale evidence is aggregated, and whether trained cell scores can concentrate fine-resolution representation. These questions are examined on two Manhattan pilot extents, one smaller and one expanded, rather than at citywide scale.

2. Study area and data

The analysis uses two Manhattan pilot extents. The smaller is a Lower Manhattan bounding box (74.02–73.97°W, 40.70–40.76°N) that yields **262** H3 resolution-9 cells (Major Revision Option B: the manuscript bbox is the analysis extent; a legacy 141-cell smoke subset is retained only for fast QA). The larger expanded-Manhattan box spans approximately 74.03–73.94°W, 40.68–40.80°N (956 R9 cells). Both lie within New York City and are described as pilot extents throughout. The H3 support follows these bounding boxes; the implications of using a rectangular support rather than a strict Manhattan land footprint are discussed in Section 5.4.

All source datasets are third-party public inputs; cell-level features, labels, fitted models, predictions, diagnostics, tables, and figures are author-derived outputs. The predictor and evidence layers are drawn from public datasets. Elevation is from the USGS 3D Elevation Program [11]; impervious fraction from the National Land Cover Database annual fractional impervious surface [12]; hydrography from NHDPlus High Resolution [13]; flood evidence from New York City Department of Environmental Protection stormwater flood polygons [14], NYC 311 street-flooding complaints [15], and USGS Hurricane Ida high-water marks [16]; and a coastal-confounding diagnostic from FEMA Sandy storm-surge inundation [17]. Machine-readable provenance is recorded in the repository download manifest (official landing pages, service URLs, SHA-256 hashes, and mirror status). DEP stormwater still resolves via a public ArcGIS FeatureServer mirror (identity not officially verified in the manifest). Street-flooding complaints are extracted from the official NYC Open Data historical dataset 76ig-c548 (2010–2014; descriptor Street Flooding (SJ); ordered pagination; unique_key deduplication; official identity verified). USGS Ida high-water marks are official (DOI 10.5066/P9OMBJPQ). Building footprints are used to compute building density. A distance-to-mapped-water proxy is derived from NHDPlus hydrography; because the Lower Manhattan hydrography is dominated by tidal rivers and shoreline features, this predictor is interpreted as shoreline/tidal-water distance rather than as inland drainage density. Polygon intersection area fractions are computed after projecting geometries to EPSG:2263. Table 1 lists the layers and their roles.

Table 1. Open data layers and their roles in the framework.

Layer	Source	Form	Role
Elevation	USGS 3DEP	Raster	Elevation, slope, flow-accumulation proxy
Impervious surface	NLCD fractional impervious	Raster	Impervious fraction
Hydrography	USGS NHDPlus HR	Vector	Shoreline/tidal-water distance proxy
Building footprints	NYC MapHub	Vector	Building density

Layer	Source	Form	Role
Flood evidence — stormwater	NYC DEP moderate-flood stormwater polygons, current sea levels (Flooding_Category 1–2)	Vector	Model-derived flood-evidence target (continuous and binary)
Flood evidence — complaints	NYC 311 street-flooding complaints (official dataset 76ig-c548, 2010–2014 Street Flooding (SJ))	Vector	Crowd-reported flood-evidence target (continuous and binary)
Flood evidence — observations	USGS Hurricane Ida high-water marks	Vector	Observed flood-evidence target (continuous and binary)
Negative control	FEMA Sandy surge inundation	Vector	Coastal-overlap diagnostic; never a training label
Held-out sensor events	NYC FloodNet events (aq7i-eu5q) + sensor metadata (kb2e-tjy3)	Vector	Strict held-out diagnostic only; never a training label
Rainfall condition r	Synthetic constant grid (75 mm/h; Supplement only)	Raster	Not used as a main claim; S_h(c) is rainfall-invariant here

The cell-level flood-evidence score combines the available open sources. Polygon sources contribute an intersection area fraction (0–1); point sources contribute a per-cell presence indicator. The score is the maximum of the area fraction and the point-presence indicator, clipped to [0,1]. The binary label is then defined deterministically as $\text{flood_class} = 1[\text{flood_evidence_score} \geq 1e-9]$, so any cell with positive evidence is **evidence-positive** and all other cells are **evidence-unrecorded**—not verified flood versus verified non-flood. Three properties of this target warrant emphasis. First, the three sources are not interchangeable measurements of one latent variable: the DEP polygons are hydrologic–hydraulic model outputs of moderate stormwater flooding at current sea levels, encoded in two ponding-depth classes (category 1, nuisance flooding with ponding depth ≥ 4 in to < 1 ft; category 2, deep-and-contiguous flooding ≥ 1 ft; the coastal "future high tides" class is excluded), the 311 complaints are crowd-reported and unverified, and only the high-water marks are direct observations of an actual event (Ida, 2021). Second, the sources span different times and forcings and are therefore collapsed into a static, "has there ever been any evidence here" score rather than a per-event inundation label. Third, the score is not a severity measure: a cell with one complaint receives the same score as a cell with a verified high-water mark, because any point presence saturates the score at 1. The score is consequently treated as a flood-evidence screening target, not as flood risk, flood depth, or a calibrated probability. Each source is retained separately in the assembled table for auditability: the DEP polygons contribute an area-fraction column (further split into nuisance and deep categories), the 311 complaints contribute count and presence fields, and the high-water marks contribute count, presence, and quality fields, with an evidence-source string recording which sources are present per cell. The composite score is then constructed from these source-specific columns rather than from an undifferentiated merge, so the claim that the sources are kept distinct is true of the assembled table, not only of the documentation. NYC 311 records in particular reflect reporting and mapping processes, and their biases have been documented previously [7,8]. NYC Open Data FloodNet street-flooding events (dataset aq7i-eu5q; published 2026-03-03) and sensor deployment metadata (kb2e-tjy3) are public and are retained for a strict held-out diagnostic. They are never joined into the composite training or evaluation target. Among study cells that contain at least one FloodNet sensor, out-of-fold scores show weak discrimination of cells with one or more quality-controlled sensor flood events (Lower Manhattan

ROC-AUC 0.314 on 23 sensor cells; expanded 0.463 on 57 sensor cells), so sensor-event agreement is not claimed as external skill. The DEP service provides the moderate-flood polygon layer under both current sea levels and a projected 2050 sea-level-rise scenario; the current-sea-level layer is taken as the primary evidence because it represents present-day flooding, while the 2050 variant is retained as a sensitivity. Re-running the target with the 2050 layer shifts Option B ($n = 262$) positive prevalence from 63.7% to 67.6% and pooled ROC-AUC from 0.848 to 0.836; in the expanded pilot the shifts are 47.9% to 50.0% and 0.882 to 0.872. These changes do not alter any substantive conclusion, so the results are not an artefact of the sea-level scenario selected for the stormwater layer.

Rainfall enters the framework only as a deferred conditioning variable. The present pilots define the primary susceptibility score as $S_h(c) = f_\theta(x_c)$ from static predictors alone; a constant synthetic rainfall input (Ida-like 75 mm/h) is retained for scenario bookkeeping in the Supplement but produces a flat within-cell response and is not a main claim. Event-conditioned $S_h(c, r)$ is left for future work.

3. Methods

The method uses the H3 hierarchy as a common spatial reference while assigning distinct roles to the resolutions used for model fitting, spatial blocking, scale diagnostics, and adaptive refinement (Fig. 1). Production feature assembly is fail-closed: if a required static layer is missing or yields NaN after zonal aggregation, the run aborts rather than substituting synthetic hash features.

3.1 H3 representation

Supervised modelling uses H3 resolution 9 (R9) as the training and evaluation support (edge length on the order of a city block in mid-latitudes). Resolution 10 (R10) provides the fine-evidence support for the scale-loss diagnostic, with hotspots subsequently rolled to R9 and R8; model fitting remains at R9 throughout. Adaptive refinement is a post-training step that replaces selected R9 cells with their resolution-11 (R11) descendants. All H3 indexing and spatial joins use longitude–latitude coordinates (EPSG:4326). Cell areas use H3 native cell-area calculations, distance-to-water uses a great-circle (haversine) distance, and the terrain derivatives are computed on the raster grid and averaged zonally over each cell. The workflow is summarised in Fig. 1.

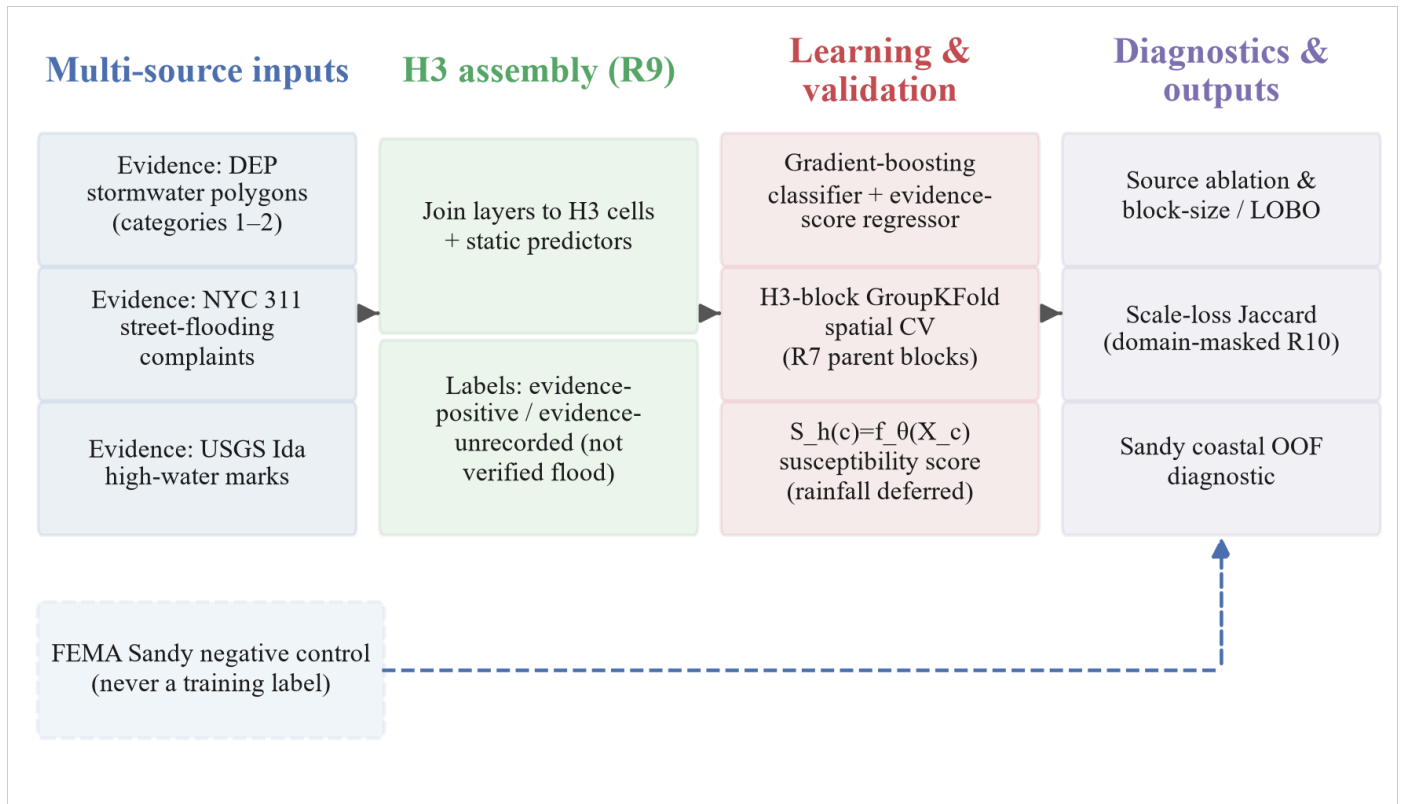


Figure 1. Open-evidence H3 pluvial-flood susceptibility workflow. Three evidence boxes—DEP stormwater polygons (categories 1–2), NYC 311 street-flooding complaints, and USGS Ida high-water marks—are assembled with static predictors into H3 R9 cells as evidence-positive / evidence-unrecorded labels. Learning yields the susceptibility score $S_h(c)=f_{\theta}(X_c)$ under H3-block spatial CV; diagnostics include source ablation, domain-masked scale-loss Jaccard, and a Sandy coastal OOF diagnostic. Rainfall conditioning is deferred. The dashed FEMA Sandy side-channel is never a training label.

3.2 Features

Static predictors are elevation, slope, a flow-accumulation proxy (D8-derived from the digital elevation model), impervious fraction (NLCD), building density, and shoreline/tidal-water distance. Elevation, slope, and the flow-accumulation proxy are zonal means over each cell derived from the digital elevation model; impervious fraction is a zonal mean of the NLCD fractional-impervious surface; building density is the building-centroid count divided by cell area; shoreline/tidal-water distance is the great-circle distance from the cell centre to the nearest NHDPlus hydrographic feature. The previously considered urban land-cover flag (impervious fraction > 0.45) is excluded from the estimator feature set because it is a deterministic copy of impervious fraction and carries no independent information. Rainfall is handled separately as the condition r rather than as a static feature.

3.3 Models and baselines

The primary learner is a gradient-boosting classifier and an evidence-score gradient-boosting regressor, each with 80 estimators, maximum depth 4, and learning rate 0.08, fitted on features standardised within each training fold, with a fixed random seed (42); all other estimator parameters retain the scikit-learn 1.8 defaults. These hyperparameters were **pre-specified without nested cross-validation tuning**. Classification uses a **prespecified operating threshold of 0.5**

for comparability across folds and pilots. Two constant classifiers—always-positive and always-negative—are computed on each held-out fold, and accuracy and F1 are reported alongside these constant classifiers to provide a class-prevalence reference. The majority-class identity is determined from pooled target counts, and the constant-baseline accuracy and F1 use the same fold-wise aggregation as the model metrics. Two further baselines are included for diagnostic comparison: an L2-regularised logistic classifier and a ponding rule defined by the weighted combination

$$0.40 \cdot (1 - \text{elev_norm}) + 0.35 \cdot \text{imperv} + 0.15 \cdot (1 - \min(\text{slope}, 15^\circ)/15^\circ) + 0.10 \cdot \text{TWI_norm},$$

where elev_norm and TWI_norm are min–max normalised elevation and a topographic-wetness proxy, and a cell is classified positive when the score is at least 0.5. In spatial cross-validation the min–max bounds for the ponding rule are computed from the training fold only and applied to the held-out fold, so the baseline does not use test-set distribution information. In-sample metrics are treated as optimistic references only. Table 2 summarises the models and baselines.

Table 2. Model and baseline specifications.

Model / baseline	Configuration	Role
Gradient-boosting classifier	80 estimators, max depth 4, learning rate 0.08, features standardised within each training fold, random seed 42	Primary binary learner
Gradient-boosting regressor	Same configuration	Evidence-score regressor
L2-regularised logistic classifier	scikit-learn 1.8 defaults	Diagnostic baseline
Ponding rule	Weighted combination of normalised elevation, impervious fraction, slope, and TWI; positive when score ≥ 0.5; bounds from training fold only	Diagnostic baseline
Always-positive / always-negative	Constant class prediction	Prevalence-aware baselines on each held-out fold

3.4 Spatial H3-block cross-validation and deployment models

Each R9 cell is assigned to its R7 parent, two H3 resolution levels coarser, and up to five-fold GroupKFold partitions these R7 groups so that no R7 block appears in both the training and the held-out portion of a fold (when fewer than five blocks are available, the number of folds equals the number of blocks). This changes the generalisation target from "new randomly drawn cells" to "new spatial blocks", which is closer to the transfer a screening product would face. For each held-out cell the predicted class probability is retained, so that threshold-independent discrimination metrics—ROC-AUC and average precision (AP), computed as the recall-weighted mean of precision across score thresholds [18]—are computed from pooled out-of-fold predictions. H3-block spatial cross-validation is the primary evaluation; random independent splits are retained as diagnostic comparisons only.

After evaluation is complete, separate deployment classifiers and regressors are refitted on all cells (fit_rows equals the pilot cell count: 262 or 956). These all-cell models are used only for descriptive maps (Fig. 2c) and adaptive-grid

screening and are never used to calculate held-out performance metrics. Model manifests record the model role (evaluation versus deployment), the number and hash of training cells, configuration, software versions, and random seed.

The R7 block size is fixed a priori; block-size sensitivity is reported in Section 4.9 and Table 8.

3.5 Scale-loss diagnostics

Fine-resolution hotspot membership is defined on native R10 evidence scores using a fixed 10% area budget with fractional allocation within boundary ties (no lexicographic H3-index tie-break and no cell-count top-k). Fine cells are restricted by a **study_domain_mask** so that R10 parents at modelling resolution R9 equal the modelling R9 set (Option B: $n_{\text{coarse}} = 262$), removing the bbox-induced parent surplus (previously 296 vs 262). For each coarse parent p the reference membership is the area-weighted fraction $w_{\text{ref}}(p) = \sum_{i \in p} a_i w_i / \sum_{i \in p} a_i$. Coarse hotspots are selected under the same area budget on area-weighted mean, maximum, or p90 parent scores, allowing fractional membership of the final tied group so that selected coarse area equals the fine budget within floating-point tolerance. The primary overlap metric is the area-weighted soft Jaccard; hard-set Jaccard is reported only as a seeded tie-resolution sensitivity (1000 draws; median and 95% interval). Continuous distortion is reported as mean absolute error between fine-parent mean scores and the coarse aggregation. Maximum aggregation can approach high Jaccard for structural reasons (parent collapse under max) and is not interpreted as a preferred strategy. Table 4 and Fig. 6b are read from the same archived diagnostics table; figures never recompute Jaccard independently. The fine R10 reference is assembled by overlaying the raw polygon and point geometries directly onto R10 cells, not by inheriting scores from R9 parents. These diagnostics use different labels, resolutions, and hotspot definitions from the Jaccard value reported by Svelling et al. [5] and are not a reproduction of that result.

3.6 Adaptive representation experiment (Supplement)

After training, the full-fit positive-class model score screens R9 cells for refinement as an **adaptive representation experiment** reported by cell count only (Supplementary Fig. S2; Table 5). A cell is selected when its score is at or above the 0.8 quantile of all cell scores, or when its predicted probability is uncertain (uncertainty $1 - 2|p - 0.5|$ of at least 0.7, i.e. p between 0.35 and 0.65); the selection is then expanded to include the one-ring H3 neighbourhood among the R9 cells. Each selected cell is replaced by its R11 descendants while unselected cells remain at R9. This step does **not** re-extract R11 features or refit a fine-resolution model; it changes representation size only. Runtime, memory, hotspot retention, and prediction quality are not claimed.

3.7 Susceptibility score (rainfall deferred)

The primary cell index is

$$S_h(c) = f_{\theta}(X_c),$$

the classifier's positive-class score from static predictors X_c (not a calibrated probability; not PF1b). Rainfall-conditioned $S_h(c, r)$ is defined for future event-based analyses but is not claimed here: the pilot's constant synthetic

rainfall has zero training variance, so flat scenario loops are retained only as a limitation/Supplement note.

3.8 Coastal-confounding diagnostic

FEMA Sandy coastal inundation is excluded from feature construction, target construction, model fitting, and model selection. It is attached only after evidence assembly and used as a coastal-confounding diagnostic rather than a strict negative control. Because the flood-evidence score combines polygon overlap and point presence, the pluvial grouping in the negative control is derived from the same composite flood_class definition used elsewhere, so that a cell carrying only a point label is counted as pluvial rather than misassigned to the "neither" group. The control reports the overlap between pluvial evidence and coastal inundation (the coastal-only fraction) and the difference in the **out-of-fold model score** between pluvial-only and coastal-only cells. The score compared is the held-out gradient-boosting score, never the target flood-evidence score, because a coastal-only cell has flood_class = 0 by construction and therefore a target score of 0 by definition; comparing target scores would be circular. Sandy labels are never used as training labels.

4. Results

4.1 Spatial pattern of evidence, predictions, and the full-fit score

On the 262-cell Lower Manhattan R9 support (Option B), the open-evidence scores, out-of-fold scores, and full-fit model score show related but distinct spatial patterns (Fig. 2). The evidence scores are bimodal by construction: cells with no positive evidence score 0, whereas positive evidence yields either a fractional polygon-overlap score or a point-presence score of 1 (median 1.0, mean 0.598). Fig. 3 shows how this composite surface decomposes into its source components: the DEP stormwater polygons and the 311 crowd reports occupy overlapping but not identical cell sets, and the composite score is the maximum over the source-specific layers. The Ida high-water marks contribute no cells within the Lower Manhattan extent—the USGS Ida points lie outside this narrow bounding box—so the Lower Manhattan composite is driven by the DEP and 311 sources alone (the expanded pilot includes 14 high-water-mark points across six cells). The out-of-fold scores are on average moderate (mean 0.663) and less dispersed, consistent with the modest pooled discrimination reported in Section 4.2 (ROC-AUC 0.850 and average precision 0.851 at 63.7% positive prevalence). The full-fit score has a mean of about 0.68 and shows moderate spatial concordance with the cross-validated surface (Pearson $r = 0.63$), as expected when the same cells, features, and labels are refitted without the five-fold holdout structure. The full-fit surface is an in-sample fit and is not a validation result; predictive performance is assessed separately from the out-of-fold metrics reported in Section 4.2.

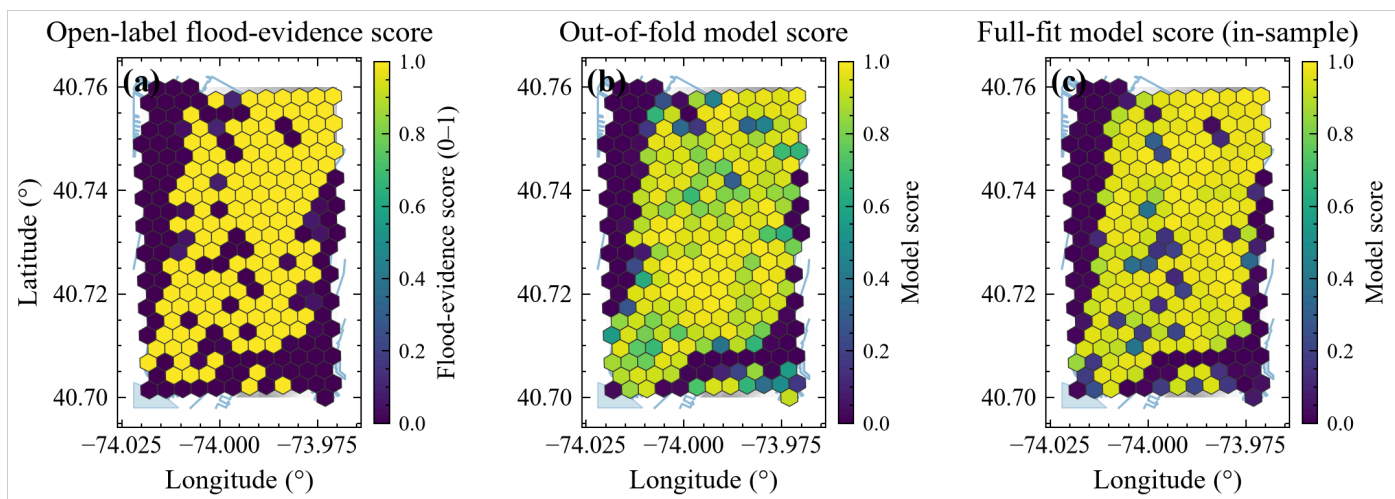


Figure 2. Spatial results for the Lower Manhattan pilot ($n = 262$ R9 cells; Option B). (a) Open-evidence flood-evidence score; (b) pooled out-of-fold gradient-boosting score under H3-block spatial cross-validation; (c) ****deployment_full**** susceptibility score $S_h(c)$, fitted using all 262 cells after evaluation. Panel (c) is an in-sample deployment fit, not a validation result.

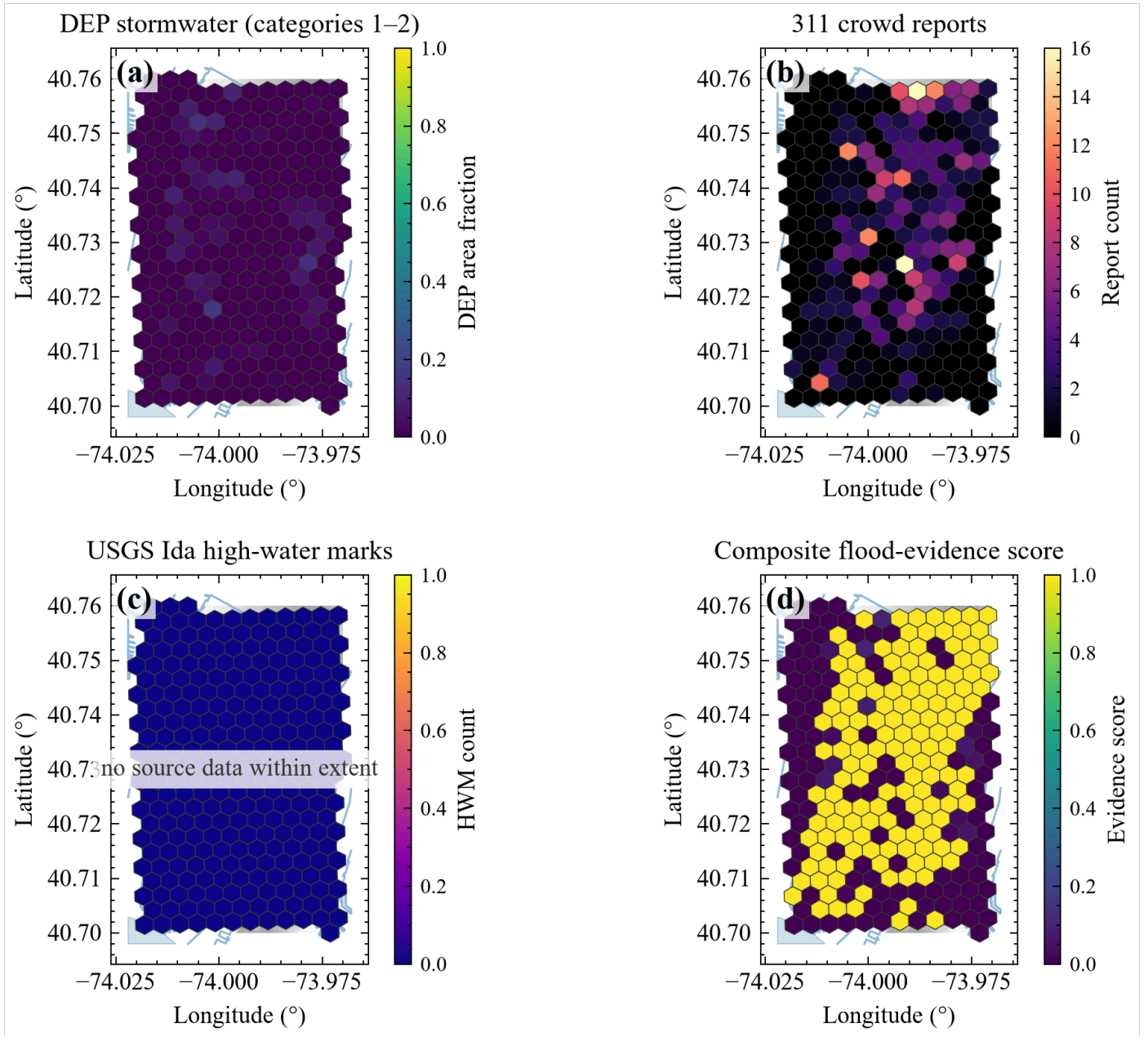


Figure 3. Source-specific open-evidence maps for the Lower Manhattan pilot ($n = 262$ R9 cells). The composite flood-evidence score (d) is the maximum over: (a) DEP stormwater polygon area fraction (categories 1–2), (b) 311 crowd-report count, and (c) USGS Ida high-water-mark count. No Ida high-water marks fall within this extent.

4.2 Spatial H3-block cross-validation

The smaller pilot contains 262 R9 cells distributed over 12 R7 blocks. Five-fold spatial cross-validation yields accuracy 0.824 ± 0.055 and F1 0.861 (Fig. 4). Here SD denotes the population standard deviation across the five held-out folds ($\text{ddof} = 0$). The held-out labels are evidence-positive in 63.7% of cells, and the model exceeds the always-positive baseline on both accuracy (0.824 vs 0.637) and F1 (0.861 vs 0.769); the always-negative baseline accuracy is 0.363. Pooled out-of-fold ROC-AUC is 0.850 and pooled average precision is 0.851, above the 0.637 prevalence reference. At the prespecified 0.5 threshold, pooled OOF balanced accuracy is 0.790, precision 0.827, recall 0.916, specificity 0.663,

and MCC 0.611. Evidence-score R^2 (0.167 ± 0.319) is demoted to a Supplement completeness metric; it quantifies fit to the constructed flood-evidence score rather than to any physical severity variable. Table 3 separates pooled OOF ranking/threshold metrics from spatial-fold variability.

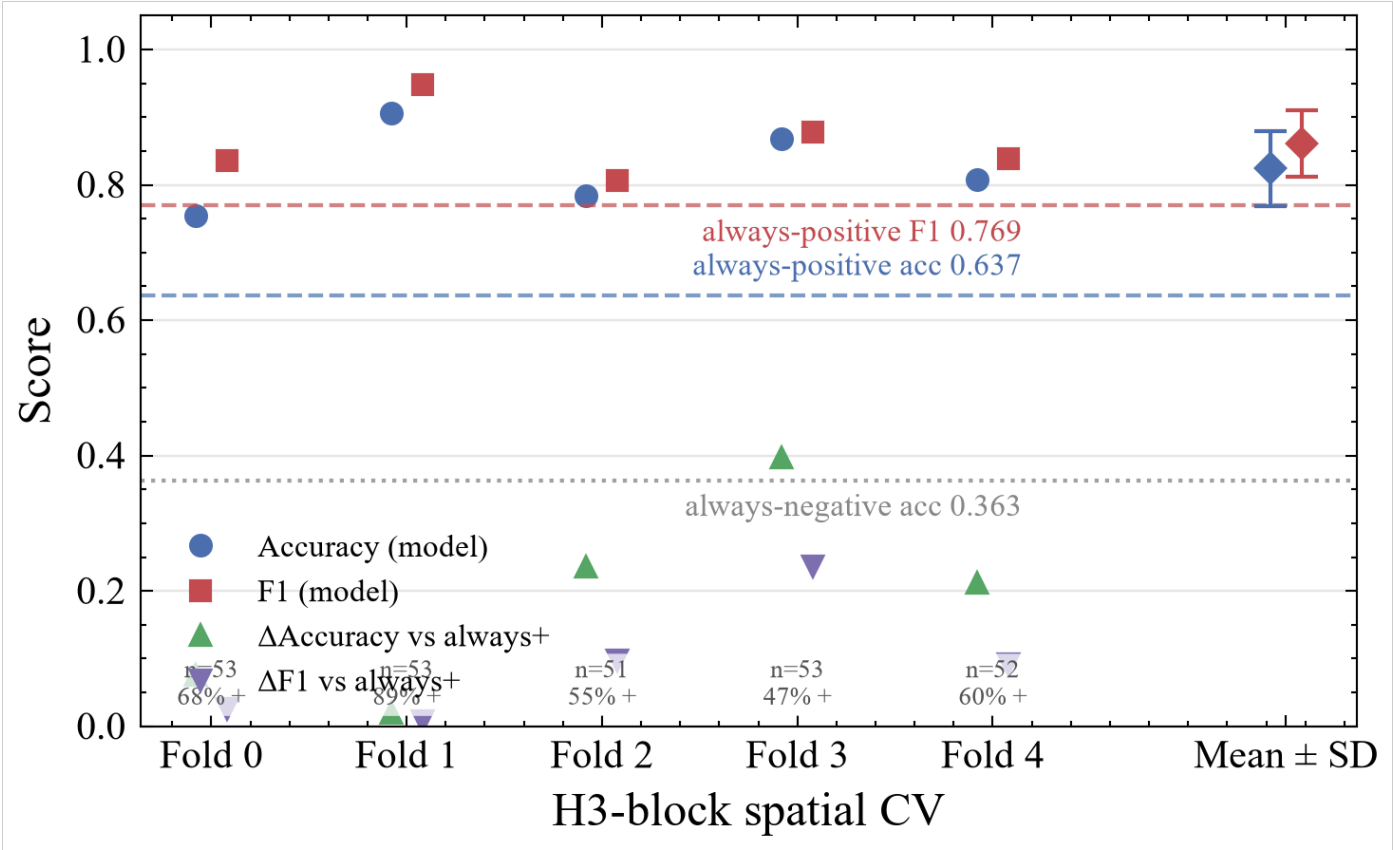


Figure 4. Spatial H3-block cross-validation performance for the Lower Manhattan pilot. Classification accuracy and F1 per fold, plus Δ Accuracy/ Δ F1 versus the fold-wise always-positive baseline; constant-class reference lines are shown. Per-fold test size and evidence-positive prevalence are annotated.

Table 3. Spatial H3-block cross-validation summary for the two pilots. Panel A: pooled out-of-fold ranking and threshold metrics (operating threshold 0.5). Panel B: fold-mean $\pm$ population SD across five held-out folds. For the smaller pilot the evidence-positive class is the majority (63.7%).

Panel A — Pooled OOF

Metric	Lower Manhattan (n = 262)	Expanded (n = 956)
ROC-AUC	0.850	0.880
Average precision	0.851	0.815
Accuracy	0.824	0.819
Balanced accuracy	0.790	0.821
Precision	0.827	0.777

Metric	Lower Manhattan (n = 262)	Expanded (n = 956)
Recall	0.916	0.876
Specificity	0.663	0.766
MCC	0.611	0.644

Panel B — Spatial-fold variability (mean $\pm$ SD)

Metric	Lower Manhattan (n = 262)	Expanded (n = 956)
Accuracy	0.824 $\pm$ 0.055	0.819 $\pm$ 0.024
F1	0.861 $\pm$ 0.049	0.822 $\pm$ 0.028
Always-positive accuracy	0.637	0.482
Always-positive F1	0.769	0.649
Always-negative accuracy	0.363	0.518

Note: SD denotes the population standard deviation across the five held-out folds (ddof = 0). Evidence-score R^2 is reported only in the Supplement.

4.3 Scale-loss Jaccard ladder

Fine-resolution hotspots are defined at R10 by a strict 10% area budget with fractional membership at tied boundary scores, after restricting fine cells to the modelling R9 study_domain_mask ($n_{\text{fine}} = 1788$; $n_{\text{coarse R9}} = 262$). Parent reference membership is area-weighted (w_{ref}), and coarse hotspots use the same area budget on mean/max/p90 scores. Under mean aggregation the R9 rollup yields area-weighted soft Jaccard 0.220 (hard median reported in Table 4) and the R8 rollup yields 0.136. Maximum aggregation reaches soft Jaccard 0.648 (R9) / 0.579 (R8); high max-aggregation Jaccard can be structural under parent collapse and is not treated as a preferred strategy. Continuous MAE between fine-parent mean scores and the coarse aggregation is 0 for mean (by construction) and 0.388 / 0.656 for max at R9 / R8. Table 4 and Fig. 6b are read from the same CSV.

Fig. 5 maps the domain-masked R10 open-evidence score surface and its mean rollups to R9 and R8, providing a spatial view of the smoothing that accompanies coarsening. Fig. 6a summarises the same scale dependence through ECDFs/histograms (better suited to 0/1-heavy scores than violins), and Fig. 6b displays the area-weighted soft Jaccard values from Table 4 (R10→R9 mean 0.220; R10→R8 mean 0.136).

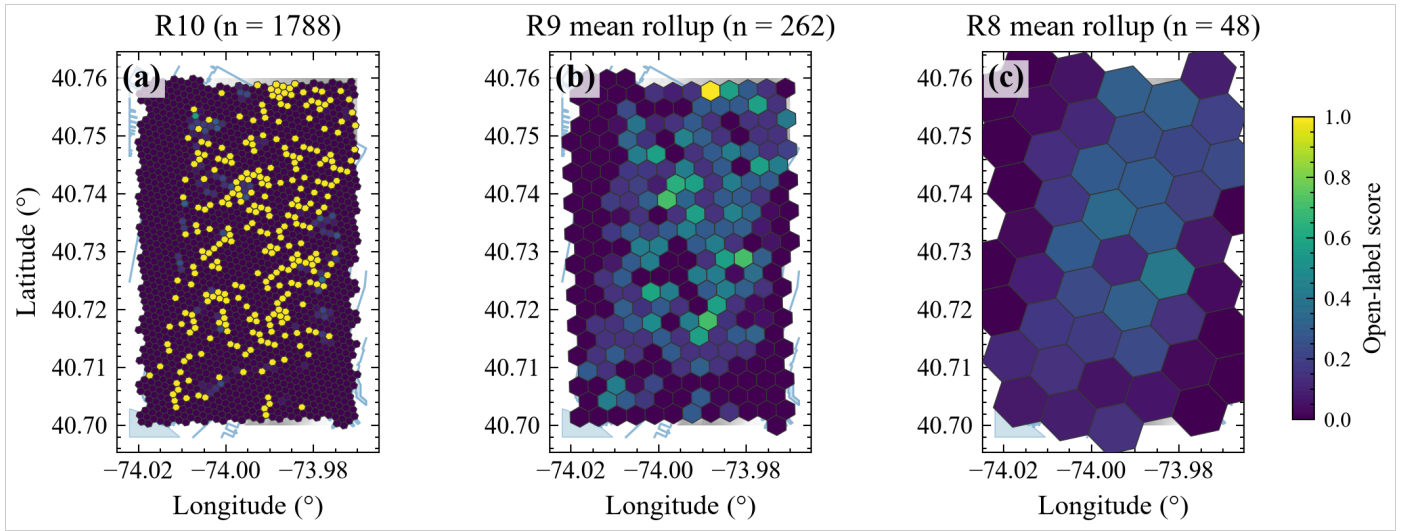


Figure 5. Multi-resolution open-evidence score surface under study_domain_mask (Option B). (a) R10 open-evidence flood score (n = 1788); (b) mean rollup to R9 (n = 262, matching modelling support); and (c) mean rollup to R8 (n = 48).

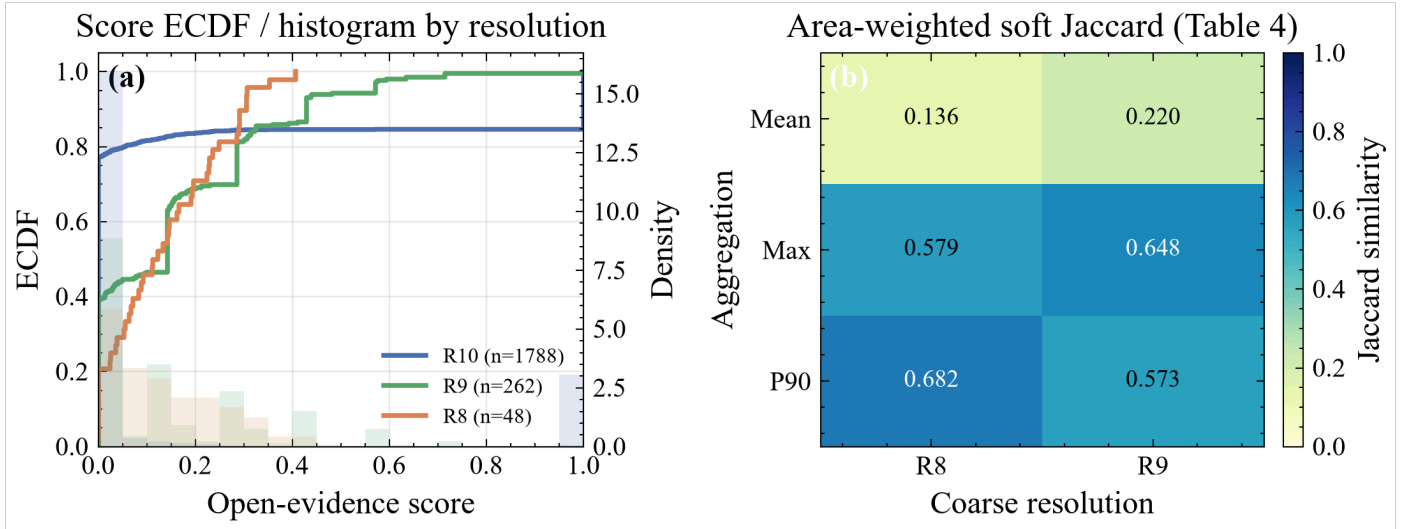


Figure 6. Resolution effects on the open-evidence score surface (Option B). (a) ECDF with light histograms of the score at R10 (n = 1788), R9 (n = 262), and R8 (n = 48); (b) area-weighted soft Jaccard from the canonical scale-results table (Table 4). For mean aggregation, R10→R9 = 0.220 and R10→R8 = 0.136.

Table 4. Scale-loss ladder under a strict 10% area budget with fractional membership ties (study_domain_mask on). Fine R10 support n = 1788 after restricting to modelling R9 parents (Option B n = 262). Primary metric is the area-weighted soft Jaccard; hard Jaccard is a 1000-draw seeded tie-resolution sensitivity. Continuous MAE is fine-parent mean versus coarse aggregation.

Coarse resolution	Aggregation	Soft Jaccard	Continuous MAE	Fine-parent recall	Coarse precision	Spearman
R9	Mean	0.220	0.000	0.360	0.360	1.000
R9	Max	0.648	0.388	0.786	0.786	0.916

Coarse resolution	Aggregation	Soft Jaccard	Continuous MAE	Fine-parent recall	Coarse precision	Spearman
R9	P90	0.573	0.251	0.729	0.729	0.983
R8	Mean	0.136	0.000	0.240	0.240	1.000
R8	Max	0.579	0.656	0.733	0.733	0.707
R8	P90	0.682	0.398	0.811	0.811	0.910

Note: cell-count budgets are not matched (area budgets are). Exact top-k budgets 5/10/15/20% remain in outputs/hotspot_budget_sensitivity.csv as a companion. The full ladder is plotted in Supplementary Fig. S1.

4.4 Adaptive representation experiment (Supplement)

The fixed coarse grid has 262 cells. Adaptive refinement selects 148 of 262 R9 cells and produces 7,366 mixed cells, compared with 12,838 cells for uniform R11 refinement (Supplementary Fig. S2). The adaptive grid therefore uses 57.4% as many cells as the uniform fine grid. True R11 feature re-extraction on refined children (post-urban-drop deployment model; 7,222 / 7,252 children scorable after dropping DEM-edge NaNs) yields hotspot recall 1.000 versus a uniformly scored scorable R11 grid (1,258 hotspots at quantile 0.9). Table 5 lists the counts.

Table 5. Adaptive representation experiment versus fixed and uniform fine grids (Lower Manhattan Option B; adaptive = 57.4% of uniform R11; 148 of 262 R9 cells refined; Option A true R11 re-extract hotspot recall 1.000).

Representation	Cell count
Fixed R9	262
Adaptive R9/R11	7,366
Uniform R11	12,838

4.5 Rainfall scenarios (limitation / Supplement)

Flat synthetic rainfall scenarios (25–100 mm/h) leave $S_{h(c)}$ unchanged within each cell. Rainfall-conditioned discrimination is **not claimed**; validated event rainfall remains future work toward $S_{h(c,r)}$.

4.6 Sandy coastal-confounding diagnostic

Across the 262 cells, 74 intersect Sandy coastal inundation and 167 carry pluvial evidence (under the composite flood_class definition); 45 have both, and 29 are coastal-only (11.1%). The control compares the **out-of-fold model score** across these groups rather than the target, because a coastal-only cell has flood_class = 0 and therefore a target score of 0 by definition; comparing target scores would be circular. The out-of-fold score averages 0.425 in coastal-only cells, 0.882 in pluvial-only cells, 0.798 in cells with both, and 0.271 in cells with neither, giving a pluvial–coastal out-of-fold score difference of 0.458. Among the 53 cells above the 0.8 score quantile, 3.8% are coastal-only and 84.9% carry pluvial evidence. As a sensitivity, excluding 311 complaints created in the Sandy landfall window 2012-10-27 to 2012-

11-05 removes 9 of 510 complaints, flips 1 cell label (167→166 positives), and leaves pooled ROC-AUC essentially unchanged (0.861 vs 0.850 on the rebuilt composite); the OOF pluvial–coastal score gap remains positive (0.438). Coastal overlap is never a training label. Table 6 details the primary overlap and mean out-of-fold scores.

Table 6. Sandy coastal-confounding diagnostic statistics on the 262-cell Option B pilot. The score comparison uses the out-of-fold model score (never the target flood-evidence score, which is 0 in coastal-only cells by construction). The pluvial–coastal difference is the mean out-of-fold score of pluvial-only cells minus that of coastal-only cells.

Statistic	Value
Cells intersecting Sandy inundation	74 of 262
Cells with pluvial evidence	167 of 262
Both pluvial and coastal	45
Coastal-only	29 (11.1%)
Pluvial-only	122 (46.6%)
Neither	66 (25.2%)
Mean out-of-fold model score, coastal-only cells	0.425
Mean out-of-fold model score, pluvial-only cells	0.882
Mean out-of-fold model score, both	0.798
Mean out-of-fold model score, neither	0.271
Pluvial – coastal out-of-fold score difference	0.458
Coastal-only among cells above 0.8 score quantile	3.8%
Pluvial among cells above 0.8 score quantile	84.9%

4.7 Expanded open-data pilot

The expanded open-data pilot contains 956 cells over 28 blocks, with positive held-out labels in 48.2% of cells (461 of 956). Under the same H3-block protocol, spatial cross-validation accuracy is 0.819 ± 0.024 and F1 is 0.822, both exceeding the fold-mean constant-class baselines (majority-negative accuracy 0.518; always-positive accuracy 0.482 and F1 0.649). Pooled out-of-fold ROC-AUC is 0.880 (fold-mean 0.880 ± 0.032) with average precision 0.815, well above the 0.482 prevalence reference; evidence-score R^2 is 0.379 ± 0.090 and MAE is 0.279. Per-fold accuracy/F1 are Fold0 0.832 / 0.848, Fold1 0.832 / 0.840, Fold2 0.827 / 0.844, Fold3 0.832 / 0.805, Fold4 0.772 / 0.776; the near-balanced per-fold class composition (positive test counts 76–102) yields a much tighter fold spread than the smaller pilot. The expanded pilot provides a within-Manhattan scale-up check rather than an independent external validation; citywide generalisation remains unevaluated.

4.8 Source-ablation of the composite target

Because the composite target is the maximum of three heterogeneous sources, its construct validity depends on whether ranking discrimination is driven by one source alone—in particular, whether the model merely re-derives the DEP hydrologic–hydraulic map from the terrain and imperviousness predictors that share its drivers. To test this, the same spatial block cross-validation was refit to DEP-only, 311-only, HWM-only, 311+HWM, and the composite; to the composite with shoreline distance removed; and to 311-only with building density removed (Table 7).

Discrimination is reproduced across source definitions rather than concentrated in DEP. In Lower Manhattan, DEP-only pooled ROC-AUC is 0.801 while 311-only reaches 0.853 (still well above 0.5) against 0.850 for the composite; in the expanded pilot the corresponding values are 0.811, 0.843, and 0.880. Dropping building density from the 311-only feature set leaves LM pooled ROC-AUC at 0.839 and expanded at 0.816—still above 0.5—so 311 skill is not an artefact of building density alone. Removing shoreline distance leaves composite discrimination essentially unchanged. The HWM-only target has no usable discrimination in these pilots.

Table 7. Source-ablation: spatial block-CV ranking discrimination for each target definition. Same protocol as the primary evaluation (H3 k=2 parent blocks, up to five folds, GBM). LM = Lower Manhattan (n = 262); Exp = expanded pilot (n = 956). These ablations reduce concern that the composite target is an artefact of a single source; they do not prove construct validity.

Target definition	Positive cells (LM / Exp)	Pooled ROC-AUC (LM / Exp)	Fold-mean ROC-AUC (LM / Exp)	F1 (LM / Exp)
DEP-only (polygon area)	74 / 231	0.801 / 0.811	0.788 / 0.814	0.507 / 0.466
311-only (crowd reports)	144 / 373	0.853 / 0.843	0.824 / 0.845	0.773 / 0.707
HWM-only (USGS Ida)	0 / 6	— / 0.233	— / 0.484	— / 0.000
311 + HWM	144 / 375	0.853 / 0.836	0.824 / 0.836	0.773 / 0.706
Composite (max)	167 / 461	0.850 / 0.880	0.834 / 0.880	0.861 / 0.822
Composite without dist_stream_m	167 / 461	0.851 / 0.887	0.825 / 0.887	0.865 / 0.831
311-only without building_density	144 / 373	0.839 / 0.816	0.821 / 0.820	0.736 / 0.667

Note: "—" marks a single-class target that was not fitted.

4.9 Block-size sensitivity and spatial autocorrelation

The R7 block size ($k = 2$) was fixed a priori, so its adequacy was tested by repeating the spatial cross-validation with R8 ($k = 1$) and R6 ($k = 3$) parent blocks, by leave-one-R7-block-out (LOBO) for Option B (12 blocks), and by computing Moran's I of the composite evidence score and of OOF residuals under native R9 k-ring adjacency (Table 8). The composite target is positively spatially autocorrelated (Moran's I 0.388 in Lower Manhattan, 0.421 in the expanded pilot), confirming that spatially blocked rather than i.i.d. splitting is required. OOF residual Moran's I is much weaker (0.048 LM; 0.108 Exp), so blocking reduces but does **not fully eliminate** residual spatial structure. Ranking discrimination is moderately stable across block sizes, yet fixed-threshold accuracy/F1 show more spatial heterogeneity—especially in the smaller pilot. LOBO on Lower Manhattan (12 folds) yields pooled ROC-AUC 0.860 (fold-mean 0.860 ± 0.137), consistent with the primary five-fold result but with wider fold dispersion when each block is held out alone.

Table 8. Block-size sensitivity of the spatial cross-validation. The same GBM protocol is repeated at parent block sizes $k = 1$ (R8), $k = 2$ (R7, primary), and $k = 3$ (R6), using **up to five folds** (when fewer blocks exist, $n_folds = n_blocks$). Moran's I is computed on the composite evidence score and on OOF residuals ($y_true - y_proba$). LM = Lower Manhattan Option B ($n = 262$); Exp = expanded pilot ($n = 956$).

Pilot	k (block resolution)	Blocks	Pooled ROC-AUC	Fold-mean ROC-AUC	Accuracy	F1
LM	1 (R8)	48	0.841	0.847	0.832	0.866
LM	2 (R7)	12	0.850	0.834	0.824	0.861
LM	3 (R6)	4	0.794	0.833	0.850	0.619
LM LOBO (R7)	2	12	0.860	0.860 ± 0.137	0.873	0.633
Exp	1 (R8)	157	0.891	0.893	0.821	0.831
Exp	2 (R7)	28	0.880	0.880	0.819	0.822
Exp	3 (R6)	7	0.872	0.851	0.799	0.780

Moran's I (composite evidence score): 0.388 (LM), 0.421 (Exp). Moran's I (OOF residual): 0.048 (LM), 0.108 (Exp). Ranking is more stable than fixed-threshold metrics; residual spatial autocorrelation is reduced but not eliminated.

5. Discussion

5.1 What the experiments establish

Taken together, the two pilots show two distinct roles of the H3 hierarchy. Spatial blocking changes the interpretation of predictive performance on **evidence-positive** / **evidence-unrecorded** labels: the Lower Manhattan model exceeds the always-positive baseline on both accuracy and F1 (0.824 vs 0.637; 0.861 vs 0.769), and the expanded pilot exceeds both the majority-negative and always-positive baselines on accuracy (0.819 vs 0.518) and F1 (0.822 vs 0.649). Coarsening under a unified study_domain_mask exposes substantial hotspot-set disagreement (mean Jaccard 0.220 to R9, 0.136 to R8 under a strict 10% area budget), while the adaptive representation experiment concentrates cells into 57% of a

uniform fine grid without any claimed efficiency gain. Pooled out-of-fold ROC-AUC indicates moderate ranking discrimination in both pilots (0.850 and 0.880). Source ablation shows that this ranking ability is not DEP-dominated: 311-only still exceeds 0.5 AUC, and 311-only without building density remains discriminative. Block-size / LOBO results show ranking is more stable than fixed-threshold metrics, and residual Moran's I remains non-zero—so spatial leakage is mitigated, not eliminated. Accordingly, the results support measurable ranking discrimination within the evaluated extents, but no citywide classification skill is claimed.

5.2 Relation to prior work

Svelling et al. [5] use H3 to aggregate a proprietary building-level index for scalable communication, whereas the present framework learns from heterogeneous open evidence and evaluates predictions across held-out H3 spatial blocks. The R8 mean-aggregation Jaccard value of 0.136 (10% area budget; domain-masked) is numerically comparable in scale to their reported value of 0.14, but the two measures use different labels, resolutions, and aggregation procedures. The present ladder therefore characterises scale loss in the open-evidence analysis rather than reproducing their metric.

5.3 Methodological implications

Methodologically, the same H3 hierarchy links the spatial unit used for evidence assembly to the units used for cross-validation, scale analysis, and subsequent refinement, extending H3 from a post-prediction visualisation layer to the learning and evaluation architecture. The framework is reproducible in the sense that every number traces to public code and public data; but reproducibility of the pipeline is not the same as validity of the label semantics, and the two are kept distinct in the accompanying audit document. The source-ablation also bears on a specific validity concern: the DEP stormwater layer is a hydrologic–hydraulic model output whose drivers (terrain, imperviousness) overlap with the predictors, so part of the composite's discrimination could in principle be model-output emulation rather than evidence fusion. That the 311-only target—an observational, crowd-reported source—still reaches pooled ROC-AUC 0.844 in the expanded pilot indicates the framework is not merely re-deriving the DEP map, even though the composite gains only a small margin over the better single source.

5.4 Limitations

Several limitations constrain the interpretation of the results. The evidence is restricted to two sub-city Manhattan extents: Lower Manhattan Option B ($n = 262$) and the expanded pilot ($n = 956$). The open evidence, particularly the 311 complaints, reflects reporting and mapping processes rather than complete ground-truth inundation [7,8], and the DEP stormwater layer is itself a hydrologic–hydraulic model output rather than an observation; only the high-water marks are direct observations. The H3 support follows rectangular bounding boxes rather than a strict Manhattan land footprint, so shoreline and water cells can shift prevalence and predictor distributions; the shoreline/tidal-water distance predictor is a tidal-water proxy rather than an inland drainage-density measure; the urban land-cover flag was removed from production features as a deterministic copy of impervious fraction. The terrain and flow-accumulation predictors are heuristic proxies computed on a georeferenced raster without hydrological conditioning, and the DEM interpolation and resolution are those of the underlying export service.

Evaluation is further constrained by class composition and the number of spatial blocks. The smaller Option B pilot is 63.7% evidence-positive and distributes 12 H3 blocks across five folds; the expanded pilot uses 28 blocks. Block-size sensitivity and LOBO (Section 4.9) show ranking discrimination is moderately stable across R8/R7/R6 blocking, while fixed-threshold accuracy/F1 are more heterogeneous; residual Moran's I (0.047–0.101) shows spatial leakage is reduced but not eliminated. Evidence-score R^2 is demoted to the Supplement. Adaptive refinement is a cell-count representation experiment only. The classifier output is not calibrated.

Rainfall conditioning remains unevaluated under observed forcing ($S_h(c)$ only). Land-mask proxies and DEP area-fraction thresholds (0/1%/5%/10%) are reported as sensitivities in the registry; a true land-fraction ≥ 0.5 polygon mask was not assembled. The urban flag remains a deterministic transform of impervious fraction. The FloodNet held-out diagnostic covers only the sparse sensor footprint and does not support citywide event skill claims.

5.5 Outstanding steps

Future work should first replace the constant rainfall placeholder with documented event observations and test $S_h(c, r)$ for fixed static features. The same spatial cross-validation protocol can then be applied over broader extents, with denser FloodNet coverage and calibrated probabilities. Spatial buffering around held-out blocks remains a priority because residual Moran's I shows leakage is not fully eliminated.

6. Conclusions

Using H3 as a common spatial support links open-evidence learning to spatially blocked validation, scale diagnostics, and selective refinement rather than treating the grid only as a post-processing layer. Across the two Manhattan pilots, the blocked evaluation and prevalence-aware baselines change the interpretation of classification performance, and selective refinement links resolution control to the fitted cell scores without requiring uniform fine-grid representation. The evidence remains limited to the evaluated pilot extents; observed event rainfall, rainfall-responsive predictions, citywide evaluation, and denser FloodNet event validation remain priorities for further assessment.

CRediT authorship contribution statement

[待补充 — to be completed before submission: list each author with their CRediT roles.]

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Declaration of competing interest

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Figure captions

Figure 1. Open-evidence H3 pluvial-flood susceptibility workflow. Three evidence boxes—DEP stormwater polygons (categories 1–2), NYC 311 street-flooding complaints, and USGS Ida high-water marks—are assembled with static predictors into H3 R9 cells as evidence-positive / evidence-unrecorded labels. Learning yields the susceptibility score $S_h(c)=f_\theta(X_c)$ under H3-block spatial CV; diagnostics include source ablation, domain-masked scale-loss Jaccard, and a Sandy coastal OOF diagnostic. Rainfall conditioning is deferred. The dashed FEMA Sandy side-channel is never a training label.

Figure 2. Spatial results for the Lower Manhattan pilot (n = 262 R9 cells; Option B). (a) Open-evidence flood-evidence score; (b) pooled out-of-fold gradient-boosting score under H3-block spatial cross-validation; (c) **deployment_full** susceptibility score $S_h(c)$, fitted using all 262 cells after evaluation. Panel (c) is an in-sample deployment fit, not a validation result.

Figure 3. Source-specific open-evidence maps for the Lower Manhattan pilot (n = 262 R9 cells). The composite flood-evidence score (d) is the maximum over: (a) DEP stormwater polygon area fraction (categories 1–2), (b) 311 crowd-report count, and (c) USGS Ida high-water-mark count. No Ida high-water marks fall within this extent.

Figure 4. Spatial H3-block cross-validation performance for the Lower Manhattan pilot. Classification accuracy and F1 per fold, plus Δ Accuracy/ Δ F1 versus the fold-wise always-positive baseline; constant-class reference lines are shown. Per-fold test size and evidence-positive prevalence are annotated.

Figure 5. Multi-resolution open-evidence score surface under study_domain_mask (Option B). (a) R10 open-evidence flood score (n = 1788); (b) mean rollup to R9 (n = 262, matching modelling support); and (c) mean rollup to R8 (n = 48).

Figure 6. Resolution effects on the open-evidence score surface (Option B). (a) ECDF with light histograms of the score at R10 (n = 1788), R9 (n = 262), and R8 (n = 48); (b) area-weighted soft Jaccard from the canonical scale-results table (Table 4). For mean aggregation, R10→R9 = 0.220 and R10→R8 = 0.136.

Supplementary Figure S1. Open-evidence hotspot scale-loss diagnostics across H3 resolutions. Reads the same CSV as Table 4 (domain-masked).

Supplementary Figure S2. Adaptive representation experiment versus uniform fine grids by cell count. Fixed R9 (262), adaptive mixed R9/R11 (7,222), and uniform R11 (12,838) for Lower Manhattan Option B (cell-count only; not runtime or R11 re-extraction).

Data and code availability

The public repository (code, configs, tests, paper documentation, and small summary tables; large rasters, GeoJSON files, trained model binaries, and large parquet files are excluded) is available at <https://github.com/Coucou2016/pluvial-flood-risk-DGGS-H3>. Provenance of the reported numeric registry and diagnostics is archived with the paper materials; each raw layer is mapped in the repository download manifest to its source URL, retrieval date, and license. Analyses that produced the frozen paper artifacts used scikit-learn 1.8.0 and H3 4.4.2 as recorded in the run metadata (created 2026-08-30); the repository pins these versions for reproducibility. Synthetic demonstrations are excluded from scientific evidence. A process-oriented research report and a data-authenticity audit accompany the manuscript.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this work, generative AI tools assisted with language editing, code review, test design, figure consistency checks, and drafting of the technical audit. The authors executed the scientific code, verified numerical outputs against regenerated artifacts, and take full responsibility for the content. Author names, affiliations, ORCID, and CRediT statements remain placeholders pending confirmation.

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